

Database [E24-Class 5]

Term project

Overview

In this project, students need to design and build a database system for a predefined topic. Your database must be implemented in one of familiar database management system, such as Oracle, MySQL, SQL Server... This project must be done in groups of two or three students. Any change in the group composition, once the groups have been formed, will need my approval.

Completion of the project requires submission of different reports at various due dates throughout the semester as well as in-class system demonstration.

The project

Phase	Description	Due date	Percentage
1	Group formation	22/8/2026	0
2	• Requirement analysis for “real-world scenario” assigned • Conceptual design • Logical design	03/10/2026	10% (group)
3	• Normalization • Data implementation	14/11/2026	10% (group)
4	Demonstration of completed system, answer questions	21/11/2026	20% (individual)

Phase 2:

In this phase, you need to finish the first 3 steps of the Database design process.

1. Identify the “real-world scenario” that you are assigned to model in your database,
2. Construct an Entity-relationship diagram and all needed description,
3. Convert the ERD into a set of relational schemas.

The report of this phase must include the following:

- All required information for the database that you are assigned, including a description of the database applications: an analysis of the data requirements that should include all of the relevant information needed to be collected, all constraints, restrictions and relationship between the data, all types of applications that can be built on top of the database (loads, updates, queries, reports, ...).
- An entity-relationship diagram (ERD) depicting your conceptual design. The ERD must capture all of the constraints that are possible using the E-R modeling concepts and notations. Any constraints or requirements of the business that cannot be modeled by the E-R notation should be clearly stated in English. Your ERD must present all types entity sets, all types of attributes, and all types of relationships.
- A set of relational schemas converted from the above ERD. All primary keys, foreign keys and connections among the relational schemas must be clearly presented.

Phase 3:

This phase includes two steps:

1. Normalization: define all functional dependencies for each relational schema. Identify all candidate keys for each table. Convert all relational schemas in previous phase into the 3rd Normal form.
2. Physical database design: deals with the implementation of the database using any type of database management systems. All tables must be populated with sufficient amount of data to generate meaningful results for various types of queries corresponding to different applications.

The report in this phase must include:

- All functional dependencies and keys for each relational schema, and all normalized relational database schema in the 3rd normal form must be clearly presented.
- Proofs of your entire database implementation. All data files and applications presented by SQL commands should be included.

Phase 4:

During this phase your group will demonstrate your database system in class.

Requirements:

1. All members of the group must show up at the time of demonstration.
2. Each member of the group must present some parts of the project as instructed by the lecturer.
3. Depending on your workload, your presentation time will be long or short.

General requirements

Each group will assign one member to send reports at corresponding due dates to my email, hoand@ptit.edu.vn. All email titles/subjects **must** include **[E24-Class 5] and your group number**, for example: [E24-Class 5] group 2 – report phase 2.

All reports are prepared in English, and must be submitted in one single file each time (MS Word, or PDF only). All other types of report will not be accepted.

Reports must be submitted by **23h59' on due dates**.

All reports will be graded in a 10-score scale. All members in each group are expected to get the same grades for group reports (phases 2 and 3). However, if the contribution to the project in the group is different among its members, the individual grades will be adjusted accordingly.